SwiftAssist – AI Customer Support Workflow

1. Problem Statement

SwiftAssist, a Tier 1 and Tier 2 food delivery startup, is experiencing a surge in customer complaints about delayed deliveries and refunds. Currently, manual support is too slow, often taking 24+ hours to resolve simple queries like refund status or order tracking. This delay is causing frustration, reduced customer satisfaction, and higher churn rates.

2. Proposed AI Workflow

Step-by-Step Process (Input → Reasoning → Action → Output):

1. **Input (Customer Query):**
   Customer sends a message like *"Where is my order?"* or *"When will I get my refund?"*
2. **Reasoning (AI Understanding):**
   - Detect intent (Order Status, Refund Check, Delivery Time, Complaint).
   - Extract entities (Order ID, payment type, delivery date).
3. **Action (Data Retrieval & Decision):**
   - Fetch relevant data from internal systems (order history, payment gateway, delivery partner status).
   - Decide whether the query can be resolved autonomously or needs escalation.
4. **Output (Response):**
   - If simple: Provide an **instant update** (e.g., "Your order will arrive in 15 minutes" or "Your refund will be credited by 3 days").
   - If complex: **Escalate to human agent** with context (AI passes conversation + order details).

Example Flow:
Customer asks "Where is my order?" → AI fetches order status → Responds with real-time tracking → Ends chat unless delay exceeds threshold, in which case escalates.

3. Example Prompts / LLM Workflows

1. **Order Status Prompt:**
   "You are SwiftAssist's AI support agent. A customer asks about their order status. Retrieve the delivery partner's live location and provide an estimated delivery time."

2. **Refund Status Prompt:**
   "You are SwiftAssist's AI agent. A customer wants to know when their refund will arrive. Check the last cancelled order, verify refund initiation, and respond with the expected refund date."

3. **Escalation Prompt:**
   "You are SwiftAssist AI. A customer is angry about a missing delivery and is requesting compensation. Escalate the case to a human support agent with all order details and customer complaint summary."

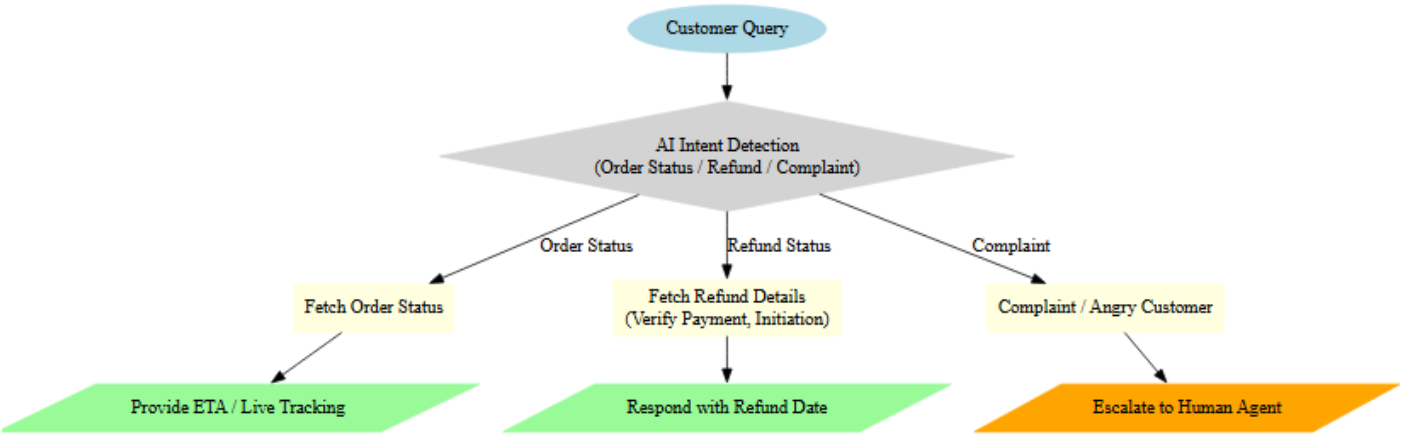
4. Key Metrics (KPIs)

- **Average Response Time:** Target < 1 minute.
- **% Queries Resolved without Human Involvement (Automation Rate):** Target 60–70%.
- **Customer Satisfaction (CSAT) Score:** Target improvement from current baseline (e.g., 3.2 → 4.5).

5. Risks and Challenges

- **Incorrect Refund Processing:** AI might mistakenly confirm refunds when not initiated.
- **Lack of Empathy in Responses:** Robotic tone could upset already frustrated customers.

Guardrails / Safeguards:

- Human approval for refunds above ₹500.
- Escalation if customer expresses high dissatisfaction (anger detection).
- Regular fine-tuning with real chat logs for natural and empathetic responses.

6. Visualization (AI Workflow Diagram Idea)